PR 2: Screenshots of the solutions

Security in Computer Networks

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Apr 23, 2026

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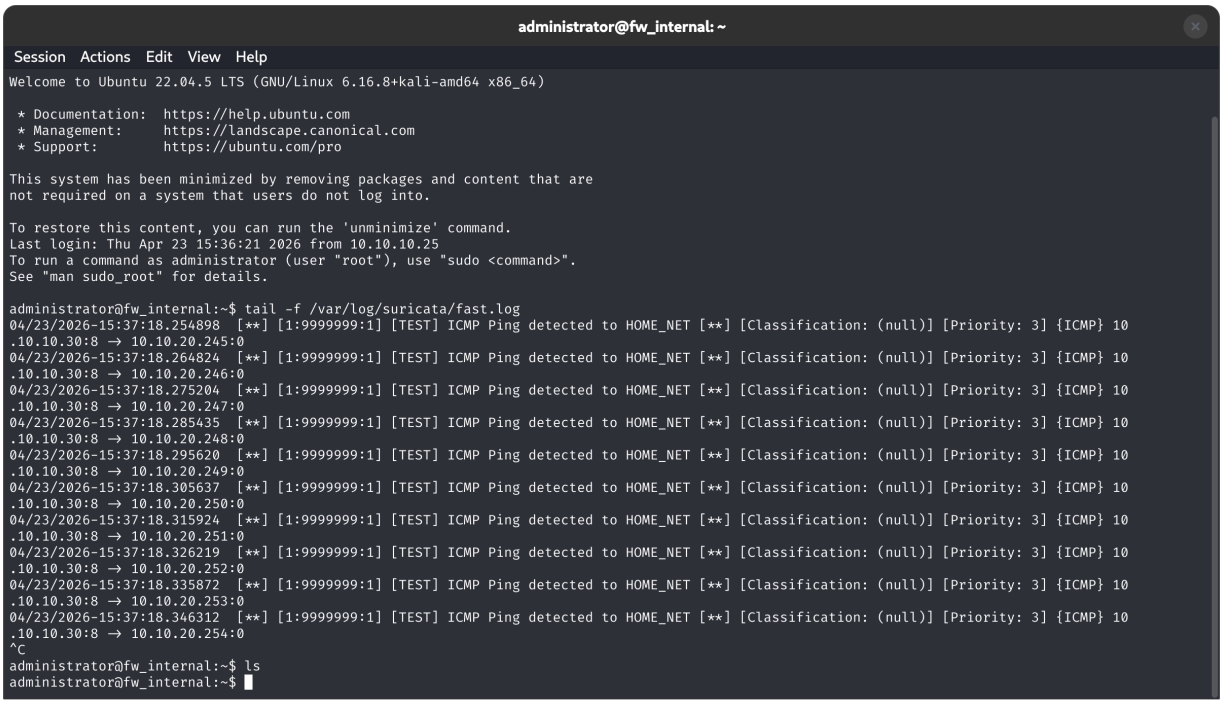
> **Note**
>
> Below are screenshots demonstrating the command results for tasks requiring them.

Task 2



```
administrator@fw_internal: ~
Session  Actions  Edit  View  Help
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.16.8+kali-amd64 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.
Last login: Thu Apr 23 15:36:21 2026 from 10.10.10.25
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

administrator@fw_internal:~$ tail -f /var/log/suricata/fast.log
04/23/2026-15:37:18.254898  [**] [1:9999999:1] [TEST] ICMP Ping detected to HOME_NET [**] [Classification: (null)] [Priority: 3] {ICMP} 10
.10.10.30:8 -> 10.10.20.245:0
04/23/2026-15:37:18.264824  [**] [1:9999999:1] [TEST] ICMP Ping detected to HOME_NET [**] [Classification: (null)] [Priority: 3] {ICMP} 10
.10.10.30:8 -> 10.10.20.246:0
04/23/2026-15:37:18.275204  [**] [1:9999999:1] [TEST] ICMP Ping detected to HOME_NET [**] [Classification: (null)] [Priority: 3] {ICMP} 10
.10.10.30:8 -> 10.10.20.247:0
04/23/2026-15:37:18.285435  [**] [1:9999999:1] [TEST] ICMP Ping detected to HOME_NET [**] [Classification: (null)] [Priority: 3] {ICMP} 10
.10.10.30:8 -> 10.10.20.248:0
04/23/2026-15:37:18.295620  [**] [1:9999999:1] [TEST] ICMP Ping detected to HOME_NET [**] [Classification: (null)] [Priority: 3] {ICMP} 10
.10.10.30:8 -> 10.10.20.249:0
04/23/2026-15:37:18.305637  [**] [1:9999999:1] [TEST] ICMP Ping detected to HOME_NET [**] [Classification: (null)] [Priority: 3] {ICMP} 10
.10.10.30:8 -> 10.10.20.250:0
04/23/2026-15:37:18.315924  [**] [1:9999999:1] [TEST] ICMP Ping detected to HOME_NET [**] [Classification: (null)] [Priority: 3] {ICMP} 10
.10.10.30:8 -> 10.10.20.251:0
04/23/2026-15:37:18.326219  [**] [1:9999999:1] [TEST] ICMP Ping detected to HOME_NET [**] [Classification: (null)] [Priority: 3] {ICMP} 10
.10.10.30:8 -> 10.10.20.252:0
04/23/2026-15:37:18.335872  [**] [1:9999999:1] [TEST] ICMP Ping detected to HOME_NET [**] [Classification: (null)] [Priority: 3] {ICMP} 10
.10.10.30:8 -> 10.10.20.253:0
04/23/2026-15:37:18.346312  [**] [1:9999999:1] [TEST] ICMP Ping detected to HOME_NET [**] [Classification: (null)] [Priority: 3] {ICMP} 10
.10.10.30:8 -> 10.10.20.254:0
^C
administrator@fw_internal:~$ ls
administrator@fw_internal:~$
```

Figure 1: Task 2 commands

Task 4

```
aperez-b@uoc.edu@kali:~$ ssh administrator@10.10.10.2 -p 40022
The authenticity of host '[10.10.10.2]:40022 ([10.10.10.2]:40022)' can't be established.
ED25519 key fingerprint is: SHA256:DdTx2BRkp5uKpHbUpoTearVO1KQe+OCrzcE/6qiArPQ
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[10.10.10.2]:40022' (ED25519) to the list of known hosts.
administrator@10.10.10.2's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.16.8+kali-amd64 x86_64)

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This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

administrator@fw_internal:~$ tail -f /var/log/suricata/fast.log
04/23/2026-16:21:29.137919  [**] [1:9999999:1] [TEST] ICMP Ping detected to HOME_NET [**] [Classification: (null)] [Priority: 3] {ICMP} 10.10.10.30
:8 -> 10.10.20.246:0
04/23/2026-16:21:29.147863  [**] [1:9999999:1] [TEST] ICMP Ping detected to HOME_NET [**] [Classification: (null)] [Priority: 3] {ICMP} 10.10.10.30
:8 -> 10.10.20.247:0
04/23/2026-16:21:29.158590  [**] [1:9999999:1] [TEST] ICMP Ping detected to HOME_NET [**] [Classification: (null)] [Priority: 3] {ICMP} 10.10.10.30
:8 -> 10.10.20.248:0
04/23/2026-16:21:29.168675  [**] [1:9999999:1] [TEST] ICMP Ping detected to HOME_NET [**] [Classification: (null)] [Priority: 3] {ICMP} 10.10.10.30
:8 -> 10.10.20.249:0
04/23/2026-16:21:29.178879  [**] [1:9999999:1] [TEST] ICMP Ping detected to HOME_NET [**] [Classification: (null)] [Priority: 3] {ICMP} 10.10.10.30
:8 -> 10.10.20.250:0
04/23/2026-16:21:29.188908  [**] [1:9999999:1] [TEST] ICMP Ping detected to HOME_NET [**] [Classification: (null)] [Priority: 3] {ICMP} 10.10.10.30
:8 -> 10.10.20.251:0
04/23/2026-16:21:29.201427  [**] [1:9999999:1] [TEST] ICMP Ping detected to HOME_NET [**] [Classification: (null)] [Priority: 3] {ICMP} 10.10.10.30
:8 -> 10.10.20.252:0
04/23/2026-16:21:29.220468  [**] [1:9999999:1] [TEST] ICMP Ping detected to HOME_NET [**] [Classification: (null)] [Priority: 3] {ICMP} 10.10.10.30
:8 -> 10.10.20.253:0
04/23/2026-16:21:29.230616  [**] [1:9999999:1] [TEST] ICMP Ping detected to HOME_NET [**] [Classification: (null)] [Priority: 3] {ICMP} 10.10.10.30
:8 -> 10.10.20.254:0
04/23/2026-16:21:31.516944  [**] [1:1000001:1] Scan detected in DMZ Subnet [**] [Classification: (null)] [Priority: 3] {TCP} 10.10.10.30:53046 -> 1
0.10.20.30:443
04/23/2026-16:22:34.896874  [**] [1:9999999:1] [TEST] ICMP Ping detected to HOME_NET [**] [Classification: (null)] [Priority: 3] {ICMP} 10.10.10.30
```

Figure 2: Task 4 commands

Task 7

```
aperez-b@uoc.edu@kali:~$ ssh administrator@10.10.10.2 -p 40022
The authenticity of host '[10.10.10.2]:40022 ([10.10.10.2]:40022)' can't be established.
ED25519 key fingerprint is: SHA256:DdTx2BRkp5uKpHbUpoTearVO1KQe+OCrzcE/6qiArPQ
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[10.10.10.2]:40022' (ED25519) to the list of known hosts.
administrator@10.10.10.2's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.16.8+kali-amd64 x86_64)

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This system has been minimized by removing packages and content that are
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applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

administrator@fw_internal:~$ tail -f /var/log/suricata/fast.log
04/23/2026-16:29:07.451701  [**] [1:1000002:1] SSH: connection attempt [**] [Classification: (null)] [Priority: 3] {TCP} 10.10.10.2:40022 -> 10.10.10.30:51684
04/23/2026-16:29:07.452794  [**] [1:1000002:1] SSH: connection attempt [**] [Classification: (null)] [Priority: 3] {TCP} 10.10.10.30:51684 -> 10.10.10.2:40022
04/23/2026-16:29:07.453889  [**] [1:1000002:1] SSH: connection attempt [**] [Classification: (null)] [Priority: 3] {TCP} 10.10.10.2:40022 -> 10.10.10.30:51684
04/23/2026-16:29:07.454133  [**] [1:1000002:1] SSH: connection attempt [**] [Classification: (null)] [Priority: 3] {TCP} 10.10.10.30:51684 -> 10.10.10.2:40022
04/23/2026-16:29:11.499890  [**] [1:1000002:1] SSH: connection attempt [**] [Classification: (null)] [Priority: 3] {TCP} 10.10.10.25:33398 -> 10.10.10.2:40022
04/23/2026-16:29:11.500310  [**] [1:1000002:1] SSH: connection attempt [**] [Classification: (null)] [Priority: 3] {TCP} 10.10.10.2:40022 -> 10.10.10.25:33398
04/23/2026-16:29:11.500632  [**] [1:1000002:1] SSH: connection attempt [**] [Classification: (null)] [Priority: 3] {TCP} 10.10.10.25:33398 -> 10.10.10.2:40022
04/23/2026-16:29:11.905847  [**] [1:1000002:1] SSH: connection attempt [**] [Classification: (null)] [Priority: 3] {TCP} 10.10.10.25:33398 -> 10.10.10.2:40022
04/23/2026-16:29:11.906417  [**] [1:1000002:1] SSH: connection attempt [**] [Classification: (null)] [Priority: 3] {TCP} 10.10.10.2:40022 -> 10.10.10.25:33398
04/23/2026-16:29:11.906678  [**] [1:1000002:1] SSH: connection attempt [**] [Classification: (null)] [Priority: 3] {TCP} 10.10.10.25:33398 -> 10.10.10.2:40022
04/23/2026-16:29:11.907580  [**] [1:1000002:1] SSH: connection attempt [**] [Classification: (null)] [Priority: 3] {TCP} 10.10.10.2:40022 -> 10.10.10.25:33398
04/23/2026-16:29:11.907592  [**] [1:1000002:1] SSH: connection attempt [**] [Classification: (null)] [Priority: 3] {TCP} 10.10.10.2:40022 -> 10.10.10.25:33398
04/23/2026-16:29:11.907972  [**] [1:1000002:1] SSH: connection attempt [**] [Classification: (null)] [Priority: 3] {TCP} 10.10.10.25:33398 -> 10.10.10.2:40022
04/23/2026-16:29:11.908118  [**] [1:1000002:1] SSH: connection attempt [**] [Classification: (null)] [Priority: 3] {TCP} 10.10.10.25:33398 -> 10.10.10.2:40022
04/23/2026-16:29:12.907873  [**] [1:1000002:1] SSH: connection attempt [**] [Classification: (null)] [Priority: 3] {TCP} 10.10.10.2:40022 -> 10.10.10.25:33398
04/23/2026-16:29:12.908731  [**] [1:1000002:1] SSH: connection attempt [**] [Classification: (null)] [Priority: 3] {TCP} 10.10.10.25:33398 -> 10.10.10.2:40022
04/23/2026-16:29:13.908086  [**] [1:1000002:1] SSH: connection attempt [**] [Classification: (null)] [Priority: 3] {TCP} 10.10.10.2:40022 -> 10.10.10.25:33398
04/23/2026-16:29:13.909045  [**] [1:1000002:1] SSH: connection attempt [**] [Classification: (null)] [Priority: 3] {TCP} 10.10.10.25:33398 -> 10.10.10.2:40022
04/23/2026-16:29:14.908268  [**] [1:1000002:1] SSH: connection attempt [**] [Classification: (null)] [Priority: 3] {TCP} 10.10.10.2:40022 -> 10.10.10.25:33398
04/23/2026-16:29:14.909118  [**] [1:1000002:1] SSH: connection attempt [**] [Classification: (null)] [Priority: 3] {TCP} 10.10.10.25:33398 -> 10.10.10.2:40022
04/23/2026-16:29:15.908480  [**] [1:1000002:1] SSH: connection attempt [**] [Classification: (null)] [Priority: 3] {TCP} 10.10.10.2:40022 -> 10.10.10.25:33398
04/23/2026-16:29:15.909283  [**] [1:1000002:1] SSH: connection attempt [**] [Classification: (null)] [Priority: 3] {TCP} 10.10.10.25:33398 -> 10.10.10.2:40022
04/23/2026-16:29:16.908583  [**] [1:1000002:1] SSH: connection attempt [**] [Classification: (null)] [Priority: 3] {TCP} 10.10.10.2:40022 -> 10.10.10.25:33398
04/23/2026-16:29:16.909674  [**] [1:1000002:1] SSH: connection attempt [**] [Classification: (null)] [Priority: 3] {TCP} 10.10.10.25:33398 -> 10.10.10.2:40022
04/23/2026-16:29:17.908838  [**] [1:1000002:1] SSH: connection attempt [**] [Classification: (null)] [Priority: 3] {TCP} 10.10.10.2:40022 -> 10.10.10.25:33398
04/23/2026-16:29:17.909169  [**] [1:1000002:1] SSH: connection attempt [**] [Classification: (null)] [Priority: 3] {TCP} 10.10.10.25:33398 -> 10.10.10.2:40022
04/23/2026-16:29:18.908953  [**] [1:1000002:1] SSH: connection attempt [**] [Classification: (null)] [Priority: 3] {TCP} 10.10.10.2:40022 -> 10.10.10.25:33398
04/23/2026-16:29:18.909326  [**] [1:1000002:1] SSH: connection attempt [**] [Classification: (null)] [Priority: 3] {TCP} 10.10.10.25:33398 -> 10.10.10.2:40022
04/23/2026-16:29:19.909095  [**] [1:1000002:1] SSH: connection attempt [**] [Classification: (null)] [Priority: 3] {TCP} 10.10.10.2:40022 -> 10.10.10.25:33398
04/23/2026-16:29:19.909483  [**] [1:1000002:1] SSH: connection attempt [**] [Classification: (null)] [Priority: 3] {TCP} 10.10.10.25:33398 -> 10.10.10.2:40022
04/23/2026-16:29:20.909229  [**] [1:1000002:1] SSH: connection attempt [**] [Classification: (null)] [Priority: 3] {TCP} 10.10.10.2:40022 -> 10.10.10.25:33398
04/23/2026-16:29:20.909519  [**] [1:1000002:1] SSH: connection attempt [**] [Classification: (null)] [Priority: 3] {TCP} 10.10.10.25:33398 -> 10.10.10.2:40022
```
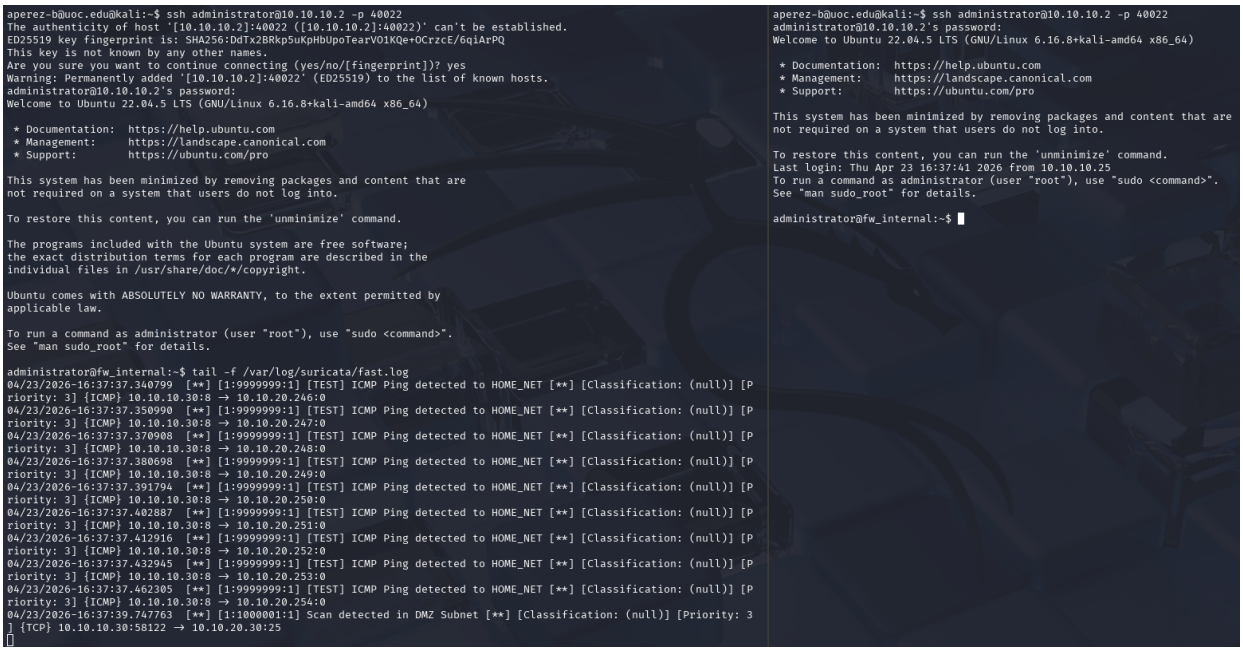

Figure 3: Task 7 commands

Task 8



```
aperez-b@uoc.edu@kali:~$ ssh administrator@10.10.10.2 -p 40022
The authenticity of host '[10.10.10.2]:40022 ([10.10.10.2]:40022)' can't be established.
ED25519 key fingerprint is: SHA256:DdTx2BRkp5uKpHbUpoTearVO1KQe+OCrzcE/6qiArPQ
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[10.10.10.2]:40022' (ED25519) to the list of known hosts.
administrator@10.10.10.2's password:
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applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

administrator@fw_internal:~$ tail -f /var/log/suricata/fast.log
04/23/2026-16:37:37.340799  [**] [1:9999999:1] [TEST] ICMP Ping detected to HOME_NET [**] [Classification: (null)] [P
riority: 3] {ICMP} 10.10.10.30:8 -> 10.10.20.246:0
04/23/2026-16:37:37.350990  [**] [1:9999999:1] [TEST] ICMP Ping detected to HOME_NET [**] [Classification: (null)] [P
riority: 3] {ICMP} 10.10.10.30:8 -> 10.10.20.247:0
04/23/2026-16:37:37.370908  [**] [1:9999999:1] [TEST] ICMP Ping detected to HOME_NET [**] [Classification: (null)] [P
riority: 3] {ICMP} 10.10.10.30:8 -> 10.10.20.248:0
04/23/2026-16:37:37.380698  [**] [1:9999999:1] [TEST] ICMP Ping detected to HOME_NET [**] [Classification: (null)] [P
riority: 3] {ICMP} 10.10.10.30:8 -> 10.10.20.249:0
04/23/2026-16:37:37.391794  [**] [1:9999999:1] [TEST] ICMP Ping detected to HOME_NET [**] [Classification: (null)] [P
riority: 3] {ICMP} 10.10.10.30:8 -> 10.10.20.250:0
04/23/2026-16:37:37.402887  [**] [1:9999999:1] [TEST] ICMP Ping detected to HOME_NET [**] [Classification: (null)] [P
riority: 3] {ICMP} 10.10.10.30:8 -> 10.10.20.251:0
04/23/2026-16:37:37.412916  [**] [1:9999999:1] [TEST] ICMP Ping detected to HOME_NET [**] [Classification: (null)] [P
riority: 3] {ICMP} 10.10.10.30:8 -> 10.10.20.252:0
04/23/2026-16:37:37.432945  [**] [1:9999999:1] [TEST] ICMP Ping detected to HOME_NET [**] [Classification: (null)] [P
riority: 3] {ICMP} 10.10.10.30:8 -> 10.10.20.253:0
04/23/2026-16:37:37.462305  [**] [1:9999999:1] [TEST] ICMP Ping detected to HOME_NET [**] [Classification: (null)] [P
riority: 3] {ICMP} 10.10.10.30:8 -> 10.10.20.254:0
04/23/2026-16:37:39.747763  [**] [1:1000001:1] Scan detected in DMZ Subnet [**] [Classification: (null)] [Priority: 3
] {TCP} 10.10.10.30:58122 -> 10.10.20.30:25
```

```
aperez-b@uoc.edu@kali:~$ ssh administrator@10.10.10.2 -p 40022
administrator@10.10.10.2's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.16.8+kali-amd64 x86_64)

 * Documentation:  https://help.ubuntu.com
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 * Support:        https://ubuntu.com/pro

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.
Last login: Thu Apr 23 16:37:41 2026 from 10.10.10.25
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

administrator@fw_internal:~$
```

Figure 4: Task 8 commands

Task 9

```
hr@web_server: ~
Session  Actions  Edit  View  Help
aperez-b@uoc.edu@kali:~$ mysql -h 10.10.20.30 -u testpr -p --disable-ssl
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 22
Server version: 8.0.45-0ubuntu0.22.04.1 (Ubuntu)

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> exit
Bye

aperez-b@uoc.edu@kali:~$ ssh hr@10.10.30.10 -p 40022
The authenticity of host '[10.10.30.10]:40022 ([10.10.30.10]:40022)' can't be established.
ED25519 key fingerprint is: SHA256:5CVVmb1k9STllgoendLItQ+jnbhXAntilOQ86eIE+5k
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[10.10.30.10]:40022' (ED25519) to the list of known hosts.
hr@10.10.30.10's password:
Permission denied, please try again.
hr@10.10.30.10's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.16.8+kali-amd64 x86_64)

 * Documentation:  https://help.ubuntu.com
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 * Support:        https://ubuntu.com/pro

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

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the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

hr@web_server:~$ mysql -h 10.10.20.30 -u testpr -p --disable-ssl
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 23
Server version: 8.0.45-0ubuntu0.22.04.1 (Ubuntu)

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> exit
Bye
hr@web_server:~$ exit
logout
Connection to 10.10.30.10 closed.

aperez-b@uoc.edu@kali:~$
```

Figure 5: Task 9 commands

Task 11

Figure 6: Task 11 commands

Task 13

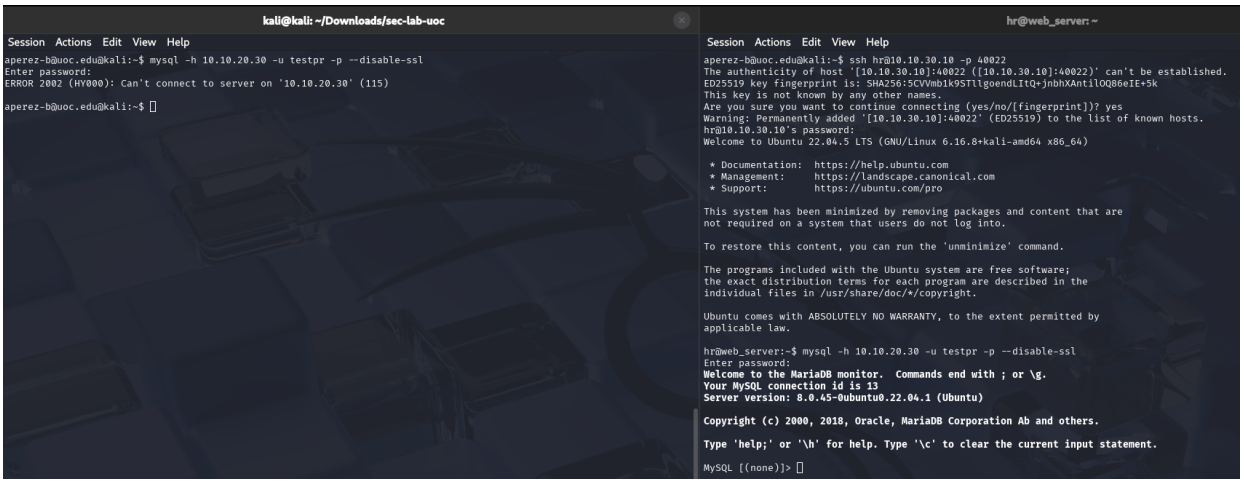

Figure 7: Task 13 commands